

MIT integration bee - 2013

Notación y Consideraciones Generales

A lo largo del documento vamos a utilizar la siguiente notación, en algunos problemas más complejos utilizaremos la notación usual:

$$\{ f(x) \}_a^b := \int_a^b f(x) dx, \quad [f(x)] := \frac{d}{dx} f(x)$$

Recursos obtenidos de: <https://math.mit.edu/~yyao1/integrationbee.html>

Herramientas que te pueden ser útiles:

- Calculadora gráfica: <https://www.desmos.com/calculator?lang=es>
- Calculadora de integrales: <https://mathdf.com/es/>
- Calculadora de integrales: <https://www.wolframalpha.com/>
- Motor de búsqueda para fórmulas en latex <https://approach0.xyz/search/>
- Te ayuda con ideas: <https://chat.deepseek.com>

Ejercicio 1

$$\int \log(x^2) - 2 \log(2x) dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \{ -\ln(4) \} \\ &= -\ln(4)x + C \end{aligned}$$

Ejercicio 2

$$\int_{-1}^3 e^{|x|} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ e^{|x|} \right\}_{-1}^0 + \left\{ e^{|x|} \right\}_0^3 \\ &= \left\{ e^{-x} \right\}_{-1}^0 + \left\{ e^x \right\}_0^3 \\ &= -\left(e^{-x} \right)_{-1}^0 + \left(e^x \right)_0^3 \\ &= e^3 + e^1 - 2 \end{aligned}$$

Ejercicio 3

$$\int \frac{(\log x)(\cos x) - (\sin x)(1/x)}{(\log x)^2} dx$$

— — — Demostración — — —

$$I = \frac{\sin(x)}{\ln(x)} + C \quad (1)$$

$$\bullet (1): \left[\frac{f(x)}{\ln(x)} \right] = \frac{f'(x) \ln(x) - f(x) \frac{1}{x}}{\ln^2(x)}$$

Ejercicio 4

$$\int_1^{11} x^3 - 3x^2 + 3x - 1 dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ (x-1)^3 \right\}_1^{11} \quad (1) \\ &= \frac{1}{4} \left((x-1)^4 \right)_1^{11} \\ &= \frac{10^4}{4} \end{aligned}$$

- (1): triángulo de pascal

Ejercicio 5

$$\int_0^2 \sqrt{12 - 3x^2} dx$$

— — — Demostración — — —

Sea $y = \sqrt{12 - 3x^2}$, entonces $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{2\sqrt{3}}\right)^2 = 1$, esta es la ecuación de una elipse con centro en $(0, 0)$, eje mayor paralelo al eje y , y radios $r_1 = 2$ $r_2 = 2\sqrt{3}$.

Sabemos que el area de una elipse es $\pi \cdot r_1 \cdot r_2$ y como la integral nos esta calculando una cuarta parte del área de la elipse tenemos que:

$$I = \frac{\pi \cdot 2 \cdot 2\sqrt{3}}{4} = \pi\sqrt{3}$$

Ejercicio 6

$$\int_0^6 x + (x-3)^7 + \sin(x-3) dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left(\frac{x^2}{2} \right)_0^6 + \left(\frac{(x-3)^8}{8} \right)_0^6 - (\cos(x-3))_0^6 \\ &= 18 \end{aligned}$$

Nota:

1. Si f es una función impar entonces $\int_{-a}^a f(x) dx = 0$

Ejercicio 7

$$\int \sin x \sqrt{1 + \tan^2 x} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \frac{\sin(x)}{\cos(x)} \right\} & (1) \\ &= -\ln |\cos(x)| + C & (2) \end{aligned}$$

- (1): $1 + \tan^2(x) = \sec^2(x)$
- (2): $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

Ejercicio 8

$$\int \frac{x^5 - x^3 + x^2 - 1}{x^4 - x^3 + x - 1} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \{ x + 1 \} \\ &= \frac{x^2}{2} + x + C \end{aligned}$$

- (1): $1 + \tan^2(x) = \sec^2(x)$
- (2): $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

Ejercicio 9

$$\int_0^1 \log x dx$$

— — — Demostración — — —

$$\begin{aligned} I &= (x \ln(x) - x)_0^1 & (1) \\ &= -1 \end{aligned}$$

$$\bullet (1): \left| \begin{array}{c} D \\ \ln(x) \\ \frac{1}{x} \end{array} \right| \left| \begin{array}{c} I \\ dx \\ x \end{array} \right|$$

Ejercicio 10

$$\int \frac{1}{1 - e^{-x}} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \frac{e^x}{e^x - 1} \right\} \\ &= \ln |e^x - 1| + C & (1) \end{aligned}$$

- (1): $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

Ejercicio 11

$$\int_0^{\pi} \sin^2 x \cos^2 x dx$$

— — — Demostración — — —

$$I = \left\{ \sin^2(\theta) \cos^2(\theta) \right\}_{-\pi/2}^{\pi/2} \quad (1), (2)$$

$$= 2 \left\{ \sin^2(\theta) \cos^2(\theta) \right\}_0^{\pi/2} \quad (3)$$

$$= \frac{1}{2} \left\{ \sin(2\theta)^2 \right\}_0^{\pi/2} \quad (4)$$

$$= \frac{1}{4} \left\{ 1 - \cos(4x) \right\}_0^{\pi/2} \quad (5)$$

$$= \frac{\pi}{8}$$

- (1): $\begin{cases} x = \pi/2 - \theta \\ dx = -d\theta \end{cases}$
- (2): $\sin(\pi/2 - x) = \cos(x)$ y $\cos(\pi/2 - x) = \sin(x)$
- (3): función par en un intervalo simétrico
- (4): $\sin(2x) = 2 \sin(x) \cos(x)$
- (5): $\cos(2x) = 1 - 2 \sin^2(x)$

Ejercicio 12

$$\int_0^{441} \frac{\pi \sin(\pi \sqrt{x})}{\sqrt{x}} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= 2\pi \left\{ \sin(u\pi) \right\}_0^{21} \quad (1) \\ &= -2 \left(\cos(u\pi) \right)_0^{21} \\ &= 4 \end{aligned}$$

- (1): $\begin{cases} t = \sqrt{x} \\ dt = \frac{1}{2} \frac{1}{\sqrt{x}} dx \end{cases}$

Ejercicio 13

$$\int \tan^2 x dx$$

— — — Demostración — — —

$$I = \left\{ \sec^2(x) - 1 \right\} \quad (1)$$

$$= \tan(x) - x + C \quad (2)$$

- (1): $1 + \tan^2(x) = \sec^2(x)$
- (2): $[\tan(x)] = \sec^2(x)$

Ejercicio 14

$$\int_0^{256} (x - \lfloor x \rfloor)^2 dx$$

— — — Demostración — — —

$$I = 256 \left\{ \text{dec}^2(x) \right\}_0^1 \quad (1)$$

$$= 256 \left\{ x^2 \right\}_0^1 \quad (2)$$

$$= \frac{256}{3}$$

- (1): la función parte decimal $\text{dec}(x) = x - \lfloor x \rfloor$ es de periodo 1
- (2): Si $x \in (0, 1)$, entonces $\text{dec}(x) = x$

Nota:

1. Sea $n \in \mathbb{Z}$ y $x \in \mathbb{R}$, entonces $\lfloor n + x \rfloor = n + \lfloor x \rfloor$

Ejercicio 15

$$\int e^{\sqrt[4]{x}} dx$$

— — — Demostración — — —

$$I = 4 \left\{ t^3 e^t \right\} \quad (1)$$

$$= 4(t^3 e^t - 3t^2 e^t + 6t e^t - 6e^t) + C \quad (2)$$

$$\bullet (1): \begin{cases} x &= t^4 \\ dx &= 4t^3 dt \end{cases}$$

$$\bullet (2): \begin{vmatrix} D & I \\ t^3 & e^t \\ 3t^2 & e^t \\ 6t & e^t \\ 6 & e^t \\ 0 & e^t \end{vmatrix}$$

Ejercicio 16

$$\int \cos x \cot x dx$$

— — — Demostración — — —

$$I = \left\{ \frac{\cos^2(x)}{\sin(x)} \right\}$$

$$= \left\{ \frac{1 - \sin^2(x)}{\sin(x)} \right\} \quad (1)$$

$$= \{ \csc(x) \} - \{ \sin(x) \}$$

$$= -\ln |\csc(x) + \cot(x)| + \cos(x) + C \quad (2), (3)$$

- (1): $\cos^2(x) + \sin^2(x) = 1$
- (2): $[\csc(x) + \cot(x)] = -\csc(x)(\csc(x) + \cot(x))$
- (3): $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

Ejercicio 17

$$\int 2 \log x + (\log x)^2 dx$$

— — — Demostración — — —

$$I = \{ (2t + t^2)e^t \} \quad (1)$$

$$= 2te^t - 2e^t + t^2e^t - 2te^t + 2e^t \quad (2)$$

$$= t^2e^t + C$$

$$\bullet (1): \begin{cases} x = e^t \\ dx = e^t dt \end{cases}$$

$$\bullet (2): \left| \begin{array}{c|c} D & I \\ \hline t^2 & e^t \\ 2t & e^t \\ 2 & e^t \\ 0 & e^t \end{array} \right|$$

Ejercicio 18

$$\int \frac{x^3}{1+x^2} dx$$

— — — Demostración — — —

$$I = \frac{1}{2} \left\{ \frac{u-1}{u} \right\} \quad (1)$$

$$= \frac{1}{2} (u - \ln|u|) + C$$

$$\bullet (1): \begin{cases} u = 1+x^2 \\ du = 2x dx \end{cases}$$

Ejercicio 19

$$\int \frac{1}{2-2x+x^2} dx$$

— — — Demostración — — —

$$I = \left\{ \frac{1}{(x-1)^2+1} \right\}$$
$$= \arctan(x-1) + C$$

Ejercicio 20

$$\int \sin x \log(\sin x) dx$$

— — — Demostración — — —

$$I = -\cos(x) \ln(\sin(x)) + \{ \cos(x) \cot(x) \} \quad (1)$$

$$= -\cos(x) \ln(\sin(x)) - \ln|\csc(x) + \cot(x)| + \cos(x) + C \quad (2)$$

$$\bullet (1): \left| \begin{array}{c} D \\ \ln(\sin(x)) \\ \frac{\cos(x)}{\sin(x)} \end{array} \right| \left| \begin{array}{c} I \\ \sin(x) \\ -\cos(x) \end{array} \right|$$

- (2): Ver ejercicio 16

Ejercicio 21

$$\int \frac{x}{1-x^4} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \frac{1}{2} \left\{ \frac{1}{1-u^2} \right\} \\ &= \frac{1}{4} \left\{ \frac{1}{1-u} + \frac{1}{1+u} \right\} \\ &= \frac{1}{4} (-\ln|1-u| + \ln|1+u|) + C \end{aligned} \quad (1)$$

$$\bullet (1): \begin{cases} t &= x^2 \\ dt &= 2x dx \end{cases}$$

Ejercicio 22

$$\int \sqrt{12-3x^2} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \sqrt{3} \left\{ \sqrt{4-x^2} \right\} \\ &= 4\sqrt{3} \left\{ \cos^2(\theta) \right\} \\ &= 2\sqrt{3} \left\{ \cos(2\theta) + 1 \right\} \\ &= 2\sqrt{3} \left(\frac{\sin(2\theta)}{2} + \theta \right) + C \\ &= 2\sqrt{3} \left(\frac{x\sqrt{4-x^2}}{4} + \arcsin(x/2) \right) + C \end{aligned} \quad \begin{matrix} (1) \\ (2) \end{matrix}$$

$$\bullet (1): \begin{cases} x &= 2 \sin(\theta) \\ dx &= 2 \cos(\theta) dx \end{cases}$$

$$\bullet (2): \cos(2x) = 2 \cos^2(x) - 1$$

Ejercicio 23

$$\int \sec^5 x \tan^3 x dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \sec^4(x)(\sec^2(x) - 1) \sec(x) \tan(x) \right\} \\ &= \left\{ u^4(u^2 - 1) \right\} \\ &= \frac{u^7}{7} + \frac{u^5}{5} + C \end{aligned} \quad (1)$$

- (1): $\begin{cases} u &= \sec(x) \\ du &= \sec(x) \tan(x) dx \end{cases}$

Ejercicio 24

$$\int_{-\pi/4}^{\pi/4} \frac{1}{1 - \sin x} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \frac{\sec(x)}{\sec(x) - \tan(x)} \right\}_{-\pi/4}^{\pi/4} \\ &= \left\{ \frac{\sec(x)(\sec(x) - \tan(x))}{(\sec(x) - \tan(x))^2} \right\}_{-\pi/4}^{\pi/4} \\ &= \left(\frac{1}{\sec(x) - \tan(x)} \right)_{-\pi/4}^{\pi/4} \quad (2), (3) \\ &= 2 \end{aligned}$$

- (1): $\begin{cases} u &= \sec(x) \\ du &= \sec(x) \tan(x) dx \end{cases}$
- (2): $[\sec(x) - \tan(x)] = \sec(x)(\tan(x) - \sec(x))$
- (3): $\left[\frac{1}{f(x)} \right] = -\frac{f'(x)}{f^2(x)}$

Ejercicio 25

$$\int \frac{1}{x\sqrt{x^2-2}} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \frac{1}{2} \left\{ \frac{1}{\frac{x}{\sqrt{2}} \sqrt{\left(\frac{x}{\sqrt{2}}\right)^2 - 1}} \right\} \\ &= \frac{\sqrt{2}}{2} \operatorname{arcsec} \left(\frac{x}{\sqrt{2}} \right) + C \end{aligned}$$

Nota:

1. Derivadas de las funciones trigonométricas inversas:

$$\begin{cases} [\arcsin(x)] &= \frac{1}{\sqrt{1-x^2}} \\ [\arccos(x)] &= \frac{-1}{\sqrt{1-x^2}} \\ [\arctan(x)] &= \frac{1}{x^2+1} \\ [\operatorname{arccot}(x)] &= \frac{-1}{x^2+1} \\ [\operatorname{arcsec}(x)] &= \frac{1}{|x|\sqrt{x^2-1}} \\ [\operatorname{arccsc}(x)] &= \frac{-1}{|x|\sqrt{x^2-1}} \end{cases}$$

